

Modelling Solar and Lunar Eclipses

Javalab Eclipse · Javalab

Year 7

~30 minutes

Science · Earth & Space Science

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Simulation: https://javalab.org/en/eclipse_en/

Curriculum links: AC9S7U03 · Earth systems

Learning intentions

- I can describe the positions of the Sun, Earth and Moon needed to cause a solar eclipse.
- I can describe the positions needed to cause a lunar eclipse.
- I can explain why eclipses do not happen every month even though the Moon orbits Earth monthly.

Getting started

Open the simulation and find the control that lets you move the Moon around its orbit. Start with the Moon in a position where no eclipse occurs, then slowly move it and watch what happens on both the diagram and the view from Earth.

Tasks

1. Move the Moon until you create a solar eclipse. Describe the relative positions of the Sun, Earth and Moon at that point.

2. Now create a lunar eclipse. Describe how the positions differ from the solar eclipse setup.

3. Which side of the Earth is in shadow during a solar eclipse, and which is in shadow during a lunar eclipse?

4. Explain, in your own words, why a solar eclipse can only happen at a particular phase of the Moon.

5. Move the Moon to three different points in its orbit and record whether each position produces a solar eclipse, a lunar eclipse, or no eclipse.

Moon's position	Eclipse type (solar/lunar/none)

6. The Moon orbits Earth roughly every month, but eclipses only happen a few times a year. Use the simulation to help explain why.

7. Describe one way this simulation's model of the Sun-Earth-Moon system also helps explain why we see different phases of the Moon.

Extension (optional)

Explain why a solar eclipse is only visible from a small area of the Earth's surface, while a lunar eclipse can be seen from anywhere on the night side of Earth.